DAY 8 — Pandas: DataFrames, Data Manipulation & Real-World Dataset Handling

Day 8 shifted focus to **Pandas**, the most popular library for data analysis. Pandas is built on top of NumPy and helps manage structured data in the form of **DataFrames**. Almost every Data Science project begins by loading a dataset using Pandas, so this day was extremely important.

🔹 1. Introduction to DataFrames

Pandas DataFrames allow:

- Row-column structure
- Easy filtering
- Fast aggregation
- Date/time handling
- Integration with CSV, Excel, SQL, JSON

We learned differences between:

- Series (1D)
- DataFrame (2D)

🔸 2. Importing & Exporting Data

We practiced:

pd.read_csv()

pd.read_excel()

df.to_csv()

Loading datasets efficiently is the first step of any ML workflow.

🔸 3. DataFrame Exploration

We revisited commands:

- df.head()
- df.tail()
- df.describe()
- df.info()

- df.shape
- df.columns

These help understand the structure of the dataset.

🔸 4. Filtering & Conditional Selection

Using:

df[df['Age'] > 30]

df[(df['City'] == 'Delhi') & (df['Salary'] > 50000)]

df.query("Marks > 80")

We practiced applying multiple conditions at once.

🔸 5. Adding, Removing & Modifying Columns

We added new derived features like:

- Percentage
- Categories
- Flags

Using:

df['new_feature'] = ...

df.drop(columns=['A'], inplace=True)

🔸 6. GroupBy & Aggregation

Understanding grouped data is extremely important.

We learned:

df.groupby('Gender').mean()

df.groupby('Department')['Salary'].max()

🔸 7. Merging & Joining DataFrames

Real-world datasets are never in one table.

We practiced:

- pd.merge()

- pd.concat()
- join()

This helped understand relational data.

🔸 8. Working with Dates

Since our major project was time series forecasting, date handling was vital.

We used:

pd.to_datetime()

df['Date'].dt.year

df['Date'].dt.month

Key Takeaways from Day 8

- Pandas makes data manipulation easy and powerful
- GroupBy and filtering are essential for EDA
- Merging/joining multiple datasets is common in real projects
- Pandas integrates deeply with ML models and visualization

This day strengthened my understanding of structured data and prepared me for time-series forecasting.